Firecracker Pollution and Regulation

GS III

- Firecrackers release smoke, gases and **fine particles** into the atmosphere during combustion. These emissions can worsen air quality, particularly when pollution from vehicles, construction and other sources is already high. Weather conditions that restrict pollutant dispersal can intensify the resulting pollution.

- **Green crackers** are designed to reduce certain emissions compared with conventional fireworks. However, reduced emissions do not mean pollution-free celebrations. When large numbers of crackers are used together, their combined emissions can still contribute significantly to deteriorating air quality.

- Environmental regulations become ineffective when restrictions are not **consistently enforced**. Illegal manufacturing, unregistered sellers and difficulty distinguishing approved products from conventional fireworks can weaken compliance. Effective regulation therefore requires reliable product identification, market monitoring and accountability throughout the supply chain.

- Air pollution control requires action across **multiple sources** rather than relying only on seasonal firecracker restrictions. Cleaner transportation, dust management, industrial emission controls and public cooperation remain important. Festival-related measures must form part of a broader strategy for improving urban air quality.

Cleaner firecrackers alone cannot prevent severe pollution. Effective air-quality protection requires enforceable standards, reduced emissions and coordinated action against multiple pollution sources.